This structure was followed to finance each US citizen's residential house during 2005-07. This boosted the real estate sector in the US because of the demand for input cost housing sector, employment opportunities in the construction sector and contributed to the US GDP. However, because of the economic cycle and massive unemployment in 2007 in the US economy, the real estate bubble burst in the US economy.. This default among the borrowers and credit protection sellers was so high that Lehman Brothers, one of the pioneer companies in the US economy, declared bankruptcy on 15th September 2008. This is because none of the stakeholders has ever anticipated that each agency will default at the same time. This crisis travelled phenomenally and was declared a global financial crisis. However, because of the conservative banking structure and with a developing financial market structure, the emerging economies were less affected by the global financial crisis..

12.4.3 How the crisis of 2007 began?

The first trouble signals were observed in early 2007 when Freddie Home Loan Mortgage Corporation FHLMC stated that it would no longer acquire high-risk mortgages and that New Century Financial Corporation, a key mortgage lender to risky customers, declared bankruptcy. Another indicator was that the ABX indexes, which track the cost of credit default insurance on instruments backed by residential mortgages, began to reflect increasing default risk assumptions about this time. Most experts agree that the crisis started in August 2007 when short-term funds were significantly removed from previously regarded securities markets. This was evidenced by sharp increases in the "haircuts" on repos and problems faced by asset-backed commercial paper (ABCP) issuers who had trouble rolling over their outstanding paper. It caused stress in the short-term funding markets in the shadow banking system during 2007 was a pervasive decline in US house prices, leading to concerns about subprime mortgages. As previously mentioned, at the start of 2007, these worries were reflected in the ABX index. In the middle of 2007, the credit rating agencies (CRAs) downgraded financial instruments backed by assets. Investors were surprised by the size of the rating actions in mid-2007 regarding the quantity of securities impacted and the average downgrading.

Most economists expected falling housing prices would devastate housing investment and consumption. Very few had forecasted that a crisis in real estate would eventually lead to a mild recession. Even fewer economists had said that this crisis might be more extensive than we think. They had anticipated that this crisis might be the biggest after the Great Depression.

When the financial crisis hit, we realised that the financial system was much more volatile than we had perceived. Within a few months, many banks and financial institutions went bankrupt or were on the verge of bankruptcy. Consequently, banks were unable or reluctant to lend because they feared they might not get their money back, given the economic situation. Investors were unwilling to invest because a lack of confidence in banking led to the failure of financial intermediaries, and credit availability reached an all-time low. As a result, the interest rate increased dramatically due to the liquidity crunch. The crisis rapidly spread to the world economy, and many banks went bankrupt. Since consumers and firms could not borrow at a very high-interest rate, consumer spending and the output level also fell.

Once the depth of the crisis was analysed, policymakers rushed to respond with financial, monetary, and fiscal policy measures. The central bank adopted a monetary policy by reducing the interest rate. The government responded by adopting expansionary fiscal policies to revive consumer demand to lift the economy out of recession. These policies certainly helped to reduce the severity of the crisis.

Over time, the central bank and the government touched on the limitation of the fiscal and monetary policies. The interest rates touched zero, and we were stuck in a "liquidity trap." Now it was the fiscal policy that was left to lift the economy out of recession. Still, with excessive expansionary fiscal policy and a significant drop in revenues due to lower output, the government had already borrowed a massive amount from the public. A considerable increase in public debt was concerning and worrisome. The government could not do enough because the public debt was too high.

It was 2010 when economic growth turned positive, but the transition to recovery was slow. The unemployment rate was high. Although emerging economies have also suffered from this crisis, large countries like India and China faced a slowdown but not a recession. Massive outflows of funds were seen from these countries because investors did not want to take any risks and were pulling the resources as much as possible to minimise the loss.

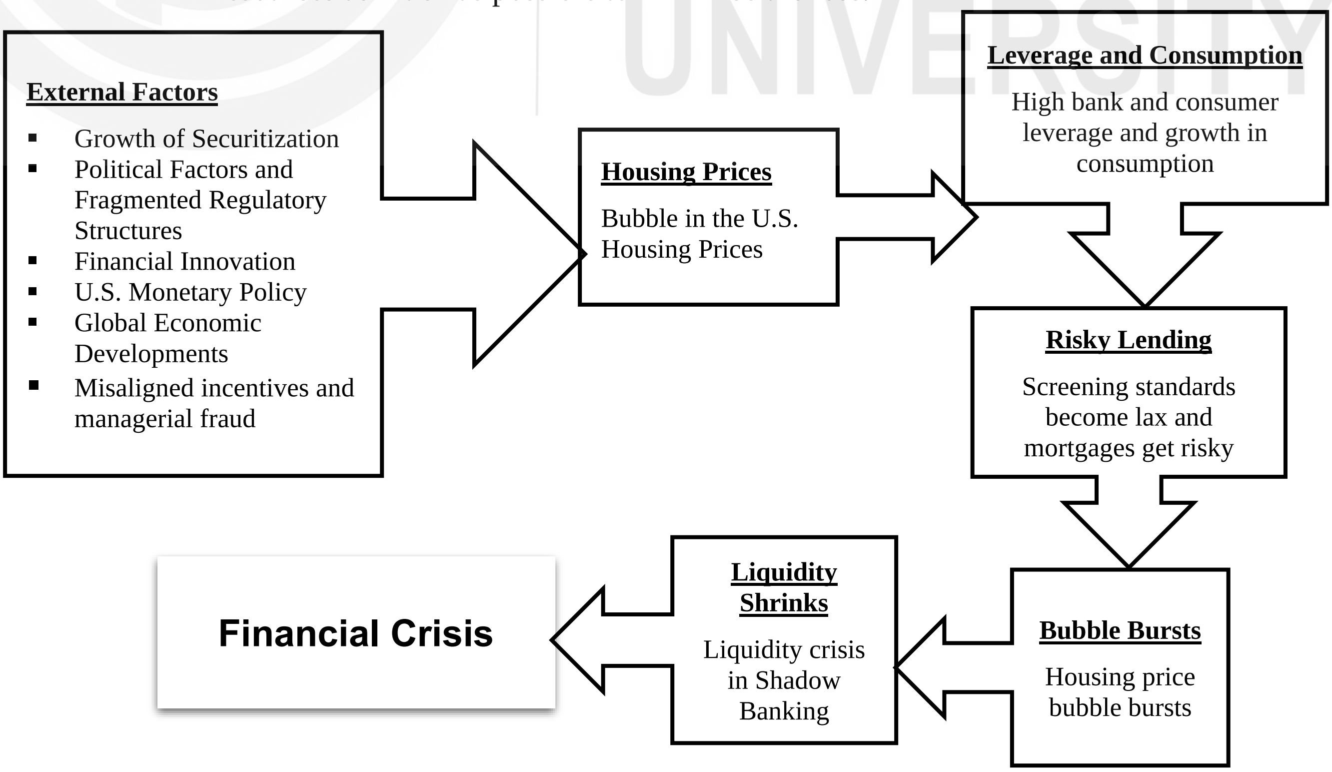

Figure 12.2 Chain of the events leading up to the crisis

12.5 WORLD'S MOST DEVASTATING FINANCIAL CRISES IN HISTORY

Financial crises have occurred throughout history and often lead to the economic tsunamis in the affected economies. Let us have a look at some major crises and their spread.

Table 12.1: Effect of various crisis all over the world

NAME OF THE CRISIS	CAUSE	ORIGIN OF THE CRISIS	AFFECTED COUNTRIES
THE CREDIT CRISIS OF 1772	This crisis was caused by a climate of overconfidence and a time of rapid loan expansion by many British banks, fueled by the vast wealth generated by their colonial holdings and trade.	London	Europe
THE GREAT DEPRESSION OF 1929-39	It was triggered by the Wall Street crash of 1929 and the poor policy decisions of the US government.	United States	Whole World
THE OPEC OIL PRICE SHOCK OF 1973	Retaliation by OPEC member countries against the US for sending arms to Israel in Arab-Israeli War abruptly halted oil exports to the US, and its allies caused significant oil shortages. A severe spike in oil prices caused an economic catastrophe in many wealthy nations, including the United States.	US and developed countries	America, Europe, South Africa, etc.
THE ASIAN CRISIS OF 1997	A period of optimism brought on by speculative capital flows from industrialised nations to the Asian Tigers led to an overextension of credit and an excessive buildup of debt in those economies., and the Thai government abandoning the fixed exchange rate against US dollars, citing a lack of foreign currency created a wave of panic across Asian financial markets.	Thailand	East Asia and Thailand trading partners
THE FINANCIAL CRISIS OF 2007-08	It was triggered by the burst of the housing bubble in the US, which threatened to bankrupt key financial institutions and enterprises.	United States	Whole World

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe the evolution of financial crises.

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2) How did the Global Financial Crisis of 2007 began?

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3) Briefly discuss the structure and the role of credit derivatives in the subprime mortgage crisis.

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12.6 THE CURRENCY CRISIS AND ITS EFFECTS ON FINANCIAL MARKETS

A currency crisis occurs when the value of a country's currency sharply declines, accompanied by volatile markets and a lack of faith in the country's economy, leading to a negative impact on the economy. As a result, exchange rates become highly unstable, and one unit of one currency may no longer be able to buy as much as it used to in another currency. Currency crises can occasionally be predicted, yet they are often sudden. Many parties can cause a currency crisis, governments, investors, central banks, or any combination of these actors.

Contrary to a currency devaluation as part of a trade war, a currency crisis is an unintended occurrence that should be avoided. Central banks and governments can help stabilise a currency by selling off foreign currency or gold reserves or by intervening in forex markets. Historically, crises have arisen when changes in investor expectations result in a substantial shift in the value of currencies.

Currency crises may be linked to other financial crises, such as abrupt halts in foreign capital inflows, significant increases in capital outflows, and defaults on sovereign debt. A currency crisis is often the symptom, not the cause, of a more severe economic downturn. Central banks are the first line of defense to safeguard a currency's stability. In a fixed exchange rate system, central banks can attempt to maintain the currently fixed exchange rate peg by drawing on the nation's foreign reserves. They can also intervene in foreign exchange markets when a potential currency crisis is present for a floating-rate system. When confidence in the stability of an economy is generally declining, investors usually try to withdraw money. This is termed as ‘capital flight’. Investors convert their domestic currency-denominated investments into foreign currency by selling them. This worsens the exchange rate, resulting in a currency run, making it nearly impossible for the country to finance its capital spending. Currency crisis prediction necessitates examining a diverse and complex set of variables. Few factors that are linked to recent crises are Current account deficit, The value of currencies rose quickly, Investor's uneasiness because of the government's measures.

The Asian Financial Crisis of 1997 is an example of a financial crisis that began as a currency crisis when Bangkok delinked the Thai baht from the US dollar; it sparked a series of currency devaluations and large-scale capital outflows.

12.7 CAUSES OF THE FINANCIAL CRISIS OF 2008

Most economists had forecast the effect of a decline in housing prices would be limited to a decrease in aggregate demand and a slowdown in growth. What they had not predicted was the impact of declining housing prices on the financial system. Sooner falling housing prices crashed the financial system as a whole. Even though it was centered on Wall Street at its peak, the global financial crisis resulted from both the US domestic and international factors.

12.7.1 US Banking system problem

Central banking's inability to force commercial banks to look out for the borrower's creditworthiness before lending and maintain a high capital ratio and low leverage ratio for transparent and robust banking led to the crisis. Commercial banks had overextended credit without accessing the borrowers' creditworthiness. And government's extensive interference and guarantees for the subprime also contributed to this crisis. For efficient and good banking, commercial banks need to maintain a good capital ratio and keep the leverage ratio minimum to show that any sudden volatility can be absorbed. But commercial banks did precisely the opposite; they did not abide by the minimum capital ratio required to maintain as per the financial regulations. If banks had maintained the high capital ratio, they wouldn't have made high returns, so they ignored the capital ratio, and because of that, the leverage ratio increased sharply. Banks probably underestimated the risks they took in good times, just like people.

It all began in the 2000s when interest was meager. Buying a house with a mortgage was easy as the mortgage rates were unusually low. The demand for housing increased significantly, which pushed up housing prices. Lenders saw it as an opportunity to profit and were willing to lend to risky borrowers aggressively. And these risky mortgages are known as subprime mortgages.

By 2006, 20 per cent of all mortgages were subprime in the US. Most economists considered this a positive development. It allowed more people to buy homes by thinking that housing prices would continue to rise relative to the house prices and they would make a profit.

In late 2006, when housing prices started to fall, many borrowers found themselves in a position where the mortgage value exceeded the house's value. It could be easily seen that the mortgages were much riskier than the lenders had assumed, and the borrowers understood. As the housing prices turned around, many borrowers started to default, and banks found themselves in a very tricky position where they were making a considerable loss. It created panic among investors, borrowers, and lenders. Investors started to sell off their financial assets, including securitised mortgages, and even went further by liquidating bank deposits. Due to liquidity shortages, even some banks began to go bankrupt. This spread had infected the other banks too. Banks were reluctant to lend even though banks were not providing liquidity to banks. The whole financial system collapsed.

12.7.2 Global Imbalances

Global trade imbalances were also significant contributors to the financial crisis. United States current account deficit with the oil-exporting countries and the Asia Pacific nations was huge during that period. Especially countries like China, Japan, and other East Asian economies had a huge trade surplus with the United States. It was not that the only United States was going through a trade deficit, but countries like Australia and New Zealand too had a considerable trade deficit.

China was the major manufacturing hub and a favorite destination for investors that helped China accumulate huge dollars reserve arising out of trade surplus and the foreign direct investment inflows. It all was facilitated by China following an undervalued and tightly pegged exchange rate. It assisted the People's Bank of China to accumulate a sizeable foreign currency reserve. Many multinational companies invested there, such as Motorola, LG, Nokia, Toshiba, etc. China has been growing at a miracle rate of 10 percent for the last two decades. By 2008 People's Bank of China had accumulated a foreign exchange reserve of more than $ US 2 trillion, most of which were government treasury bonds. China and other Asian countries' export-led growth created a huge current account deficit with the developed economies. Asian countries have a huge trade surplus with the United States, and not letting the appreciation of exchange rates lead to substantial capital outflows from the Asian economies to the US economies. And huge capital inflows to the US economy sharply led to a fall in interest rates.

Such low-interest rates made commercial banks consider parking the fund in the US housing sector. Growing China and falling interest rates in the US and global markets can be easily understood by the global loanable fund's framework. A rise in global saving shifted the saving supply schedule to the right and pushed the United States and World interest rates down.

Extremely low interest rates in the United States incentivised commercial banks to invest in housing. They started to lend without doing a risk assessment, which led to the housing crisis, and the crisis in the housing crisis caused the financial system failure and affected the macroeconomy.

12.7.3 Growth of Securitization and the OTD model

It has been asserted that the US government's intention to broaden ownership was coupled with monetary policy that allowed banks to have softer lending rules. Additionally, the originate-to-distribute (OTD) paradigm of securitisation, a lax monetary policy, and weak supervision over bank capital all contribute to this softening. Due to these circumstances, commercial banks increased their mortgage lending during the time leading up to the crisis, while investment banks engaged in warehouse lending via nonbank mortgage lenders. Additionally, there is empirical proof that the OTD model incentivised banks to make risky loans in ever-increasing numbers.

12.7.4 Financial Innovation

For more than two decades before the financial crisis, we saw an explosion of financial innovation. Information technology played a role, making it more straightforward for banks to create tradable instruments and increasing the interconnectedness of commercial banks with the shadow banking system and financial markets. But apart from information technology, there had to be economic incentives for banks to innovate. Financial markets are incredibly competitive, making it challenging for institutions to have significant profit margins with standard financial products—those whose payout distributions everyone agrees upon. This promotes the hunt for new financial products, particularly ones with contested creditworthiness. The absence of agreement on the investment value of new financial products reduces competition in that market. It enables the institutions supplying those products to make significant initial profits. But since these new items don't have a history, they are also risky. The explanation is that there is a lack of historical data that can be relied upon to resolve the conflict, which means that not only rivals but also financiers of the institutions offering these goods may disagree if they are investments worth making. When this occurs, short-term funding for innovative institutions is not rolled over, resulting in a funding crisis. Before the crisis, the explosion of new asset-backed securities created by securitisation created an ideal environment for this.

12.7.5 US Monetary Policy

Taylor (2009) argues that the price boom and subsequent crash that preceded the crisis greatly influenced the US Federal Reserve's easy-money monetary policy, particularly in the six or seven years before the crisis. He provides evidence that, between 2007 and 2009, monetary policy was too "loose fitting" in that actual interest rate decisions were significantly lower than what historical experience would indicate should be the basis for policy under the Taylor rule. Taylor (2009) demonstrates how the Federal Reserve's deliberate monetary policy choice, which included these abnormally low-interest rates, fueled the housing boom and ultimately caused the housing crisis. The study uses regression to calculate the empirical relationship between interest rates and housing starts. The results demonstrate a robust positive correlation between the intertemporal decrease in interest rates between 2001 and 2007 and the housing market boom. The impact of low-interest rates on housing prices was amplified by the incentives provided by the low-interest rate environment for lenders to make riskier loans. When the central bank keeps interest rates on loans low for an extended period, it reduces banks' net interest margins, and one way for banks to respond is to increase these margins by taking on more risk. This led banks to expand the borrower pool by lending high-risk borrowers who had previously been excluded, further fueling the housing price boom.

12.7.6 Political Factors

Rajan (2010) argues that structural flaws in the educational system that resulted in unequal access for diverse societal sectors are to blame for the widening economic inequality in the United States. Politicians from both parties saw the expansion of home ownership to deal with this growing wealth inequality—a political proclivity that dates back at least to the 19th century Homestead Act and therefore adopted legislative initiatives and other inducements to force banks to extend mortgage loans to a wider borrower base by relaxing underwriting standards, which resulted in riskier mortgage lending. The elevated demand for houses pushed house prices and led to the housing price bubble.

12.8 THE EFFECTS OF THE CRISIS ON THE MACROECONOMY

The financial crisis immediately affected the macroeconomy two folds. First, the interest rate , at which firms and people could borrow rose a lot. Usually, firms with AAA ratings borrow at a rate close to the government because the AAA rating is considered safest, and bonds with BBB ratings borrow at a higher interest rate. Still, the difference is slight, which is close to 1%. The difference in borrowing rates between the government bonds and the AAA and BBB bonds increased significantly in 2008. Borrowing rates on the government remained very low, whereas the borrowing rate on the AAA and BBB were high. This jump in interest rate made it difficult for the firms to borrow fund. It also became difficult for ordinary people to get a loan because loan default damaged their credit score, and banks were no longer willing to take risks.

Second, the September 2008 financial crisis triggered confusion among consumers and firms. People were confused and anxious about future events and were speculating about another Great Depression. This anxiety among the consumers and the firms led to a drop in confidence. Lower consumer confidence led to lower housing and stock prices, which resulted in a sharp decrease in aggregate demand. The real-world implications of this financial crisis were substantial. These included weaker household loan demand and supply, lower corporate investment, and more unemployment.

12.8.1 Credit Demand Effects

When the housing bubble burst, it jolted household balance sheets, eroding the net worth of households. As a result, the highly leveraged households cut back on their consumption. However, because of various economic frictions related to nominal price rigidities and a lower bound of zero on nominal interest rates, relatively unlevered households did not increase consumption to offset this decline.

The consumption fell sharply in places with higher reliance on home as a source of wealth and in high-leverage counties with more considerable house price reductions. Consequently, consumer spending and credit demand also decreased.

12.8.2 Credit Supply Effects

There is persuasive empirical evidence that the crisis prompted a considerable decrease in the amount of credit banks were willing to extend. One indication is the drop in syndicated loans during the crisis, which is significant because syndicated lending is an essential source of credit for the corporate sector. In addition to banks, institutional investors, hedge funds, mutual funds, insurance companies, and pension funds are also included in the syndicated loan market. The syndicated lending started to decline in the middle of 2007 and accelerated in September 2008. An increase in the cost of credit accompanied the decline in lending volume.

12.8.3 Reduction in Corporate Investment and Increase in Unemployment

It is not unexpected that corporate investment declined and unemployment rose due to a decline in consumer consumption and a rise in the cost and scarcity of credit. In 2008 and 2009, about nine million jobs were lost in the United States, accounting for around 6% of the workforce as the country descended into a deep recession. Additionally, it prevented many from attempting to return to employment after the crisis subsided, which caused the labour participation rate to plummet. This had the effect of making subsequent calculations of the unemployment rate tend to underestimate the actual jobless rate. The recorded unemployment rate increased every month from 6.2% in September 2008 to 7.6% in January 2009. By mid-2009, the average housing price in the United States had decreased by 30%, and the value of stocks had been reduced by almost 50%. The American auto sector also took significant damage. October 2008 car sales were down 31.9% from September 2008 figures. A causal link between the reduction in credit supply during the crisis and an increase in unemployment. Evidence shows that household access to bank loans tended to matter more than firm access in determining the decline in employment in the manufacturing sector. Still, there was also a significant impact from decreased access to commercial and industrial loans and consumer instalment loans.

12.9 INITIAL POLICY RESPONSE

12.9.1 Monetary policy

The central bank went for overall facilitated liquidity to the financial system to contain the damage. Given the uncertainty, federal deposit insurance was created to prevent confused depositors from rushing to withdraw the deposits. During the crisis, if the investors wanted to take their funds back, the banks had no choice but to sell some of the assets; even sometimes, the bank had to sell at a fire rate, and in the worst scenario, banks went bankrupt. To prevent this, the central bank incorporated several liquidity facilities to meet the bank's short-term fund needs. It allowed the banks to borrow from the major banks. Financial institutions were permitted to use collateral to borrow from the fed.

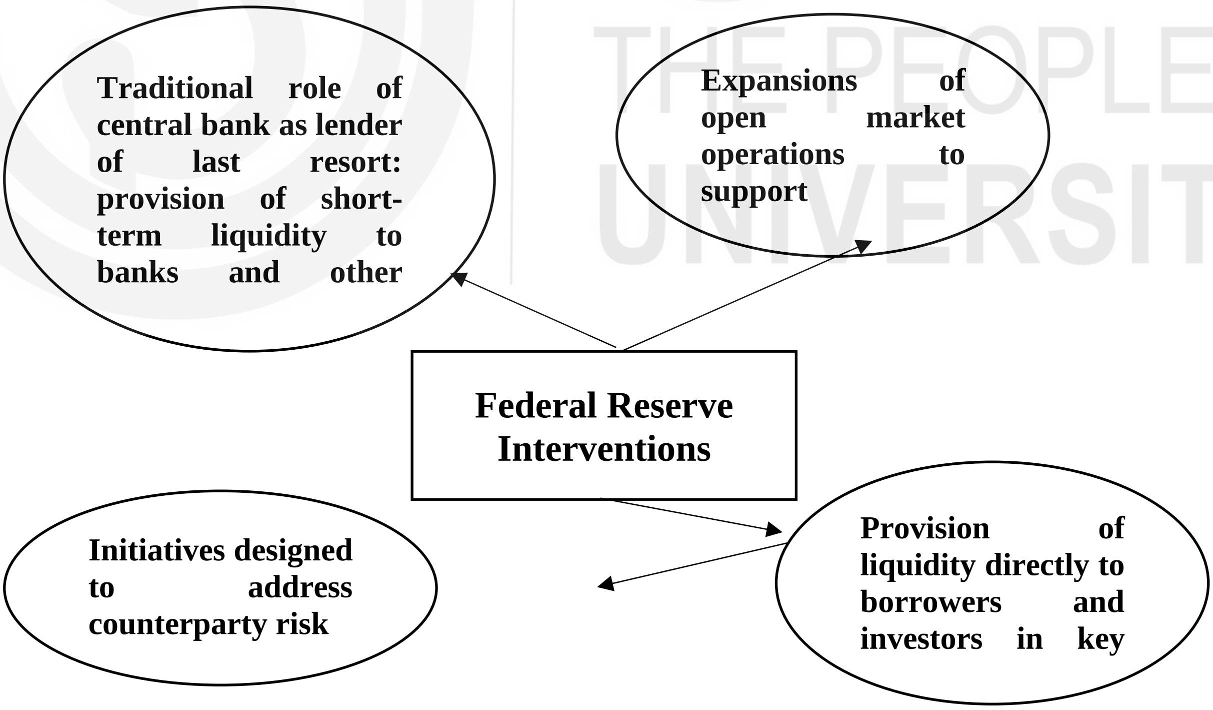

Fig. 12.3: Federal Reserve Interventions

The central banks' facilities provide much-needed liquidity to the financial system. The central banks implemented an expansionary Monetary policy to lower the interest rates so consumers and firms could easily access much-needed liquidity. The monetary authority started to restructure the bank's bad debts, so a good bank balance sheet brings investors' confidence into the system. During last months of 2009, interest rates reached near zero, and interest rates could not go down anymore. We are stuck in a liquidity trap.

12.9.2 Fiscal policy

Initially, the government introduced the Troubled Asset Relief Program (TARP), which mainly focused on cleaning up the banks. This program's initial target was 4700 billion and was introduced in October 2008. The primary purpose of TARP was to remove the bad assets from the bank's balance sheets, thus reducing uncertainty, reassuring the investors, and making the banks' balance sheets healthier. Accessing the value of impaired assets was extremely tough and time-consuming; they realised it and abandoned the original goal. The new goal was to increase the banks' capital and reduce the leverage ratio. The government started to acquire the bank's shares and provide funds to most of the largest banks to avoid the banks going bankrupt.

Once the adverse shock of the crisis became apparent, the US government adopted an aggressive expansionary fiscal policy. The government reduced taxes and increased spending. After assuming office, the Obama administration prioritised a fiscal program that would increase the demand and reduce the size of the recession. The name of that fiscal program was American Recovery and Reinvestment Act, passed in February 2009. This fiscal program package was $780 billion in the form of both tax reduction and spending. The government aggressive expansionary fiscal policy caused a large US budget deficit. The deficit increased from 1.7% of GDP in 2007 to 9% in 2010.

Despite the extensive use of a combination of monetary and fiscal policy, output decreased vastly. The economy recovered, but the recovery was slow.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Enlist the causes behind the global financial crisis of 2008.

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2) Explain the macroeconomic effects of the global financial crisis.

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